

Potent antimutagenic activity of white tea in comparison with green tea in the Salmonella assay.

There is growing interest in the potential health benefits of tea, including the antimutagenic properties. Four varieties of white tea, which represent the least processed form of tea, were shown to have marked antimutagenic activity in the Salmonella assay, particularly in the presence of S9. The most active of these teas, Exotica China white tea, was significantly more effective than Premium green tea (Dragonwell special grade) against 2-amino-3-methylimidazo[4,5-f]quinoline (IQ) and four other heterocyclic amine mutagens, namely 2-amino-3,8-dimethylimidazo[4,5-f]quinoxaline (MeIQx), 2-amino-3,4,8-trimethyl-3H-imidazo[4,5-f]quinoxaline (4,8-DiMeIQx), 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP), and 3-amino-1-methyl-5H-pyrido[4,3-b]indole (Trp-P-2). Mechanism studies were performed using rat liver S9 in assays for methoxyresorufin O-demethylase (MROD), a marker for the enzyme cytochrome P4501A2 that activates heterocyclic amines, as well as Salmonella assays with the direct-acting mutagen 2-hydroxyamino-3-methylimidazo[4,5-f]quinoline (N-hydroxy-IQ). White tea at low concentrations in the assay inhibited MROD activity, and attenuated the mutagenic activity of N-hydroxy-IQ in the absence of S9. Nine of the major constituents found in green tea also were detected in white tea, including high levels of epigallocatechin-3-gallate (EGCG) and several other polyphenols. When these major constituents were mixed to produce "artificial" teas, according to their relative levels in white and green teas, the complete tea exhibited higher antimutagenic potency compared with the corresponding artificial tea. The results suggest that the greater inhibitory potency of white versus green tea in the Salmonella assay might be related to the relative levels of the nine major constituents, perhaps acting synergistically with other (minor) constituents, to inhibit mutagen activation as well as "scavenging" the reactive intermediate(s).